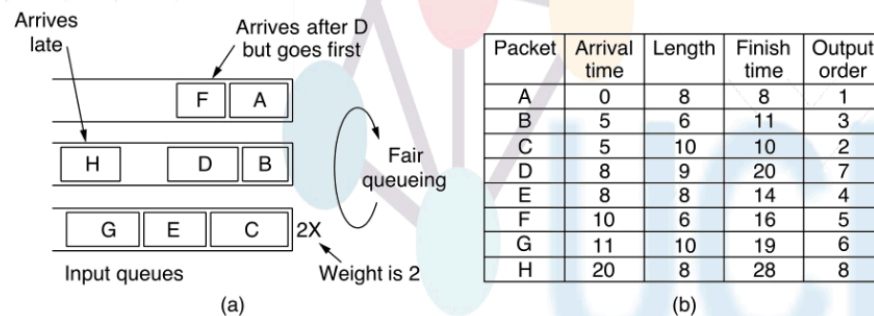


Queueing exercise

- Consider the packet queues shown in Fig. 5-29
- What is the finish time and output order of the packets if the **middle** queue, instead of the bottom queue, has a weight of 2?
- Order packets with the same finish time alphabetically



06/12/24

Figure 5-29. (a) Weighted Fair Queueing. (b) Finishing times for the packets.

41

El del medio ahora tendra x2

Packet	Arrival time	Length	Finish time	Output order
A	0	8	8	1
B	5	6	8	2
C	5	10	15	4
D	8	9	12.5	3
E	8	8	23	6
F	10	6	16	5
G	11	10	33	8
H	20	8	24	7

A computer on a 6-Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 1 Mbps. It is initially filled to capacity with 8 megabits. How long can the computer transmit at the full 6 Mbps?

Token fill rate = 1 Mbps

Bucket capacity = 8 megabits

Transmission rate = 6

El bucket se recarga mientras se envia, se recarga a un 1 Mbps por lo que el Drain Rate es de:

Drain rate = 6 Mbps - 1 Mbps = 5 Mbps

Entonces el bucket primero saca 8 - 5 Mbps, ahí se tarda un segundo, quedando 3 Mb en el bucket, por lo que el bucket quiere sacar 3 pero al sacar de 5 en 5 cada segundo saca mas de 3. Por lo que tarda menos de un segundo sacando 3 por lo que se hace $\frac{3}{5} = 0.6$ Mbps y se le suma al segundo que ya contamos.

Quedando así: Se duran 1.6 segundos en transmitir 8 megabits